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**Kanyamanan: A Web-Based Culinary Itinerary Optimization, Health Informatics,
and Restaurant Aggregator System for the Province of Pampanga**

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CHAPTER I: THE PROBLEM AND ITS SETTING

1.1 Introduction

In the contemporary global tourism and hospitality sectors, the evolution of experiential travel has positioned culinary tourism as a primary driver of regional economic growth and cultural preservation (World Tourism Organization [UNWTO], 2023). Modern travelers increasingly reject generic tourist itineraries in favor of highly localized, authentic gastronomic experiences (Hall & Gössling, 2022). This structural shift has forced the food and beverage industry to transition from static online listings toward integrated, data-driven digital ecosystems that connect independent culinary merchants into a unified, informative operational grid. Regionally, the rapid expansion of the digital economy across Southeast Asia has highlighted the necessity for localized web frameworks capable of consolidating multi-vendor data, solving real-time logistical constraints, and optimizing traveler movement across broader geographic bounds (Gavalas et al., 2022). Within the Philippine micro-, small-, and medium-enterprise (MSME) sector, there is a critical requirement for collaborative digital platforms that aggregate decentralized food heritage data into an enterprise-grade routing and wellness-tracking infrastructure to enhance market visibility without imposing complex technical onboarding barriers or operational overhead on local business owners (Melnick et al., 2022).

As the undisputed "Culinary Capital of the Philippines," the Province of Pampanga represents a vast, high-velocity gastronomic landscape where a dense concentration of heritage restaurants, ancestral kitchens, and modern food hubs span across multiple municipalities and component cities. Known for its deeply rooted culinary traditions, the province attracts massive influxes of weekend food tourists, international travelers, and transit commuters navigating major regional thoroughfares like the North Luzon Expressway (NLEX), MacArthur Highway, and the Jose Abad Santos Avenue corridors. However, this widespread provincial demand introduces severe logistical friction for visiting

consumers. Tourists attempting multi-town culinary tours frequently face uncoordinated, exhausting travel routes, unpredictable dining queues, and unoptimized travel schedules across different towns. This structural challenge directly aligns with the Personalized Tourist Trip Design Problem (PTTDP), where travel itineraries must dynamically balance geographic distance with multiple time-dependent stop constraints (Marpaung et al., 2026). Simultaneously, because these scattered culinary establishments operate as isolated digital entities, the province's food ecosystem lacks a centralized aggregator mechanism capable of balancing real-time regional traveler routing with the nutritional insights and operating schedules of individual dining venues.

Despite the widespread use of generic travel finders, mapping services, and dietary tracking applications, a significant technological gap remains in software designed to manage wide-scale, multi-vendor culinary ecosystems through intelligent data aggregation. Standard directory applications are engineered for standalone search queries and completely lack the spatial mapping logic required to calculate multi-stop, cross-municipality travel itineraries based on shifting venue operational hours, localized crowd density patterns, and individual dietary preferences (Ahmad & Halim, 2023). Furthermore, traditional platforms fail to provide a unified framework capable of handling cross-vendor data synchronization, automated nutritional item breakdown, or vision-based food metric estimation across independent culinary options (Thierry et al., 2021). This technological disconnect forces culinary travelers to rely on manual, error-prone coordination methods, leading to severe traffic congestion around popular local dining hubs and an unoptimized, frustrating experience trying to navigate Pampanga's food map manually.

To resolve these operational, spatial, and data-onboarding structural deficiencies, this study introduces Kanyamanan: A Web-Based Culinary Itinerary Optimization, Health Informatics, and Restaurant Aggregator System for the Province of Pampanga. Moving beyond a basic restaurant finder, this project is engineered

as a research-grade Decision Support System designed to dynamically coordinate, optimize, and map multi-vendor culinary exploration through a centralized data-ingestion framework. To eliminate merchant operational burdens, the platform deploys a pre-seeded data model populated by public regional tourism listings, allowing the system to be fully functional for tourists out of the box without requiring manual merchant sign-ups, live dashboard management. The architecture consists of a robust administrative portal for data curation, an optional claimable web portal for verified restaurant owners to manage their public data, and a highly responsive consumer web application optimized for desktop and mobile browsers. By integrating custom regional spatial routing logic, predictive crowd density metrics, computer vision-based nutrient analysis (Ando et al., 2021), and an automated dietary budgeting engine, Kanyamanan establishes a highly defensible, realistic, and scalable software model for modern web engineering and intelligent data aggregation in high-density culinary tourism environments.

1.2 Project Context

The system conceptualization of Kanyamanan stems from an investigation into how culinary tourism is managed in Pampanga. The existing process consists of fragmented, self-directed travel planning where tourists rely on isolated data sources. Currently, this practice is performed by manually cross-referencing blog posts, checking standalone social media pages for business hours, using generic mapping software to manually pin addresses, and tracking caloric intake through separate wellness apps. This unintegrated approach introduces severe technical, spatial, and operational inefficiencies. Spatially, generic mapping platforms completely lack the multi-stop routing logic required to cross-examine overlapping merchant operational hours against regional transit delays (Gavalas et al., 2022). Operationally, decentralization means data remains unsynchronized; dynamic factors such as seasonal menu cost changes, local crowd congestion, and dish allergen contents are hidden from travelers. Consequently, three core groups are heavily affected: culinary tourists who

endure exhausting, overlapping transit lines and long queues; local culinary MSMEs who remain digitally invisible due to technical onboarding barriers; and provincial tourism officers who lack centralized, anonymized aggregate metrics to monitor geographic foot traffic trends and optimize future development. Real-world evidence of these operational gaps is regularly observed during weekend holiday surges, where major arterial highways experience massive traffic standstills concentrated around popular local dining hubs. Consumers regularly encounter unexpected restaurant closures, long waiting lines, and menu pricing discrepancies due to outdated public information. A technology-based web intervention is necessary to consolidate these decentralized elements into a unified, informative grid. By aggregating public registries into an open operational framework, software engineering can mitigate regional transit friction, bridge the multi-vendor data gap, and provide automated health metrics without adding technical overhead to local merchants (Fang et al., 2022).

Empirical assessments of regional travel patterns show that a substantial portion of culinary tourists visiting Pampanga are same-day excursionists from adjacent metropolitan regions. Unlike overnight travelers who navigate flexible temporal windows, these day-trippers are strictly bound by sub-24-hour schedules, financial cumulative costs, and the fast physical satiety thresholds associated with the rich, high-density profiles of traditional Kapampangan cuisine. Because existing commercial mapping applications treat food trails as boundless checklists rather than constrained journeys, travelers face logistical over-scheduling, cognitive fatigue, and severe gridlock around major hubs. A specialized web system must therefore offer automated constraint scheduling tailored to these specific localized boundaries.

Statement of the Problem

The main question this study seeks to address is: How can the logistical friction, data fragmentation, and lack of system integration across the decentralized

culinary tourism ecosystem of Pampanga be resolved through an open-access web infrastructure?

Specifically, this study aims to answer the following research questions:

- How can the platform resolve the digital isolation of traditional heritage restaurants and ancestral kitchens that currently operate as separate digital entities lacking an aggregator platform for multi-vendor data synchronization?
- What technical interventions are required to mitigate the high levels of logistical transit friction encountered by domestic and international culinary travelers due to manual, error-prone route coordination across separate provincial municipalities?
- How can custom spatial routing algorithms be designed to dynamically calibrate travel sequences and destination time windows based on shifting establishment operational schedules and real-time congestion trends along major regional transit corridors?
- In what manner can the system eliminate the high technical onboarding barriers and daily operational dashboard overhead that traditionally prevent local culinary MSMEs from maintaining an active web presence?
- How can a unified framework be engineered to effectively execute automated menu item ingredient breakdowns, macro-budget tracking, allergen flagging, and mobile browser-accessible, vision-based food metric estimations across diverse dining venues?

1.3 United Nations Sustainable Development Goals (SDGs)

The development and deployment of Kanyamanan directly support global sustainability initiatives, aligning with the institutional research priorities of the University of the Assumption by leveraging software engineering to address socio-economic, health, and cultural heritage challenges.

SDG 3: Good Health and Well-being (Target 3.4)

The platform integrates advanced health informatics to promote long-term well-being and dietary mindfulness among mobile consumers. By deploying a computer vision-based calorie and macronutrient estimator alongside an automated menu ingredient deconstructor, Kanyamanan allows users to process food photographs instantly through a responsive web interface to check estimated nutritional indices against personalized health targets. The system actively flags potential allergen barriers, such as shrimp paste or nuts common in traditional Kapampangan cooking, and identifies structural dietary indicators like high saturated fats. This data-driven health modeling ensures that multi-stop culinary tours remain balanced, supporting user health and well-being directly in the field.

SDG 8: Decent Work and Economic Growth (Target 8.9)

The project establishes a digital framework designed to drive sustainable tourism and boost the market visibility of decentralized culinary MSMEs without creating financial or operational strain. Through its pre-seeded aggregator engine populated via public registries, the system integrates ancestral kitchens and small food hubs into a provincial routing matrix automatically. Local business owners do not need to possess advanced technical literacy, complete mandatory registration, or manage complex dashboards to be discovered by incoming travelers. Furthermore, the optional claimable web portal provides verified merchants with access to anonymous aggregate tourist footprint maps, allowing them to leverage data analytics to support economic growth and operational planning.

SDG 9: Industry, Innovation, and Infrastructure (Target 9.b)

The development and deployment of Kanyamanan directly support SDG 9 by pioneering localized digital innovation and modernizing the technological infrastructure available to the regional hospitality and micro-, small-, and medium-enterprise (MSME) sectors. By constructing an open-access web architecture driven by advanced cloud processing, algorithmic decision support systems, and automated data ingestion, the project establishes a resilient digital public infrastructure. This setup actively drives technological upgrading and domestic innovation within Pampanga's traditional heritage food spaces. It elevates the digital capabilities of local ancestral kitchens and independent venues without demanding complex technical literacy, upfront operational overhead, or expensive native hardware infrastructures from small-scale business owners.

SDG 11: Sustainable Cities and Communities (Target 11.4)

Kapampangan culinary traditions represent a vital component of intangible cultural heritage. Kanyamanan serves as a resilient digital public infrastructure that reinforces the safeguarding and preservation of this distinct legacy. By organizing and categorizing decentralized culinary data by municipality, the system diverts consumer traffic away from over-congested, generic commercial areas and routes travelers intentionally toward historical culinary institutions, ancestral kitchens, and localized food spaces. The platform's integrated highway traffic adjuster and predictive crowd forecasting modules help balance regional consumer distribution, protecting local heritage communities from urban congestion while sustaining the local food heritage ecosystem.

1.4 Purpose and Description

The primary purpose of this capstone project is to engineer Kanyamanan: an open-access, research-grade Decision Support System that addresses data fragmentation and logistical bottlenecks within the culinary tourism sector of Pampanga. The system operates exclusively within a web environment,

accessible across desktop and mobile web browsers without requiring the installation of native applications. Rather than executing active table bookings or handling point-of-sale transactions, the software coordinates multi-vendor navigation, dietary tracking, and regional traffic mapping through predictive constraints and standardized time-slot logic. The technical architecture is divided into three distinct operational interfaces:

Centralized Data-Ingestion & Admin Dashboard

This backend administrative interface allows system administrators to curate master records compiled from public tourism registries. The portal features database tools to pre-seed, edit, and partition establishment profiles across various municipalities. Administrators can maintain baseline data structures containing verified geographic coordinates, standard operating schedules, historical peak occupancy trends, menu price points, and dish nutritional profiles, ensuring the application functions fully out of the box.

Claimable Merchant Web Portal

This optional dashboard provides secure access for verified restaurant owners who choose to "claim" their pre-seeded public profiles. Once verified, merchants gain access to private configuration settings to verify profile accuracy and manually update digital menu lists. To assist business operations, the portal renders visual representations of anonymous, aggregated user itinerary footprint maps, allowing merchants to study regional foot traffic trends without requiring day-to-day transaction logging.

Responsive Consumer Web Application

Designed for mobile and desktop browsers, this interface allows domestic and international tourists to discover local food establishments seamlessly. Users can build optimized cross-municipality itineraries that calculate the most efficient travel sequence based on selected dining locations. The client application integrates real-time highway traffic adjustments along major transit corridors,

renders predictive crowd occupancy charts to prevent queue delays, estimates total trip expenses, tracks personal daily caloric targets, and features a computer vision interface that estimates nutritional metrics from uploaded food photos.

1.5 Objectives

The general objective of this study is to design, develop, and evaluate *Kanyamanan*, a web-based culinary itinerary optimization, health informatics, and restaurant aggregator system for the Province of Pampanga. The system serves as a browser-accessible Decision Support System configured to coordinate multi-vendor exploration, automate travel sequence optimization, and deliver health informatics to improve regional tourism efficiency and boost local merchant visibility.

Specifically, the project aims to accomplish the following technical milestones:

1. Design, develop, and implement the core *Kanyamanan* web-based platform architecture as a unified culinary itinerary optimization, health informatics, and multi-vendor restaurant aggregator infrastructure for the Province of Pampanga.
2. Develop a Provincial Food Trip Web Planner and Algorithmic Routing Engine seamlessly coupled with an adaptive Highway Traffic Adjuster module and a Predictive Crowd Density Forecaster analytical framework.
3. Construct an intelligent Health Informatics core consisting of an automated Menu Ingredient Deconstructor, an automated dietary macro budgeting engine, and a mobile-browser-accessible Computer Vision Calorie & Macronutrient Estimator.
4. Implement Intelligent Web-Integrated Assistant Modules composed of a preference-based, heritage-focused Personal Taste Matcher and a natural language Interactive Travel Assistant Chat interface, supported by frontend Progressive Web App (PWA) Offline Caching routines.

5. Evaluate the technical quality, performance efficiency, and overall user acceptability of the complete integrated web platform across diverse multi-device desktop and mobile web environments in strict accordance with the criteria and quality sub-characteristics defined by the ISO/IEC 25010 Software Quality Model.

1.6 Scope and Delimitation

Scope

The project encompasses the design, development, implementation, testing, and evaluation of a web aggregator, nutrition tracking, and itinerary planning platform configured for the culinary tourism sector of Pampanga, accessible across desktop and mobile web browsers. The system architecture follows a three-tier model composed of a centralized data-ingestion backend for administrators, an optional claimable web portal for verified merchant profile management, and a highly responsive client application for consumer access.

The geographical boundary of the study is strictly restricted to verified culinary establishments, ancestral kitchens, and modern food hubs operating within the administrative boundaries of the Province of Pampanga, focusing heavily on routing coordination along major transit corridors including NLEX, MacArthur Highway, and the Jose Abad Santos Avenue lines.

Functional modules are restricted to:

- The pre-seeded multi-vendor database engine.
- The algorithmic multi-stop trip planner featuring dynamic parameter bounding that adjusts for highway traffic trends, establishment operational windows, and an enforceable two-to-three restaurant ceiling option tailored for same-day excursionists.
- Historical peak crowd forecasting charts.

- Mobile browser-accessible computer vision models for food image analysis.
- Menu ingredient parsing routines for dietary allergen identification.
- The natural language travel chat assistant module.
- Progressive Web App (PWA) client-side caching for offline map retrieval during transit.

The system will undergo technical quality evaluation and user acceptance testing with local test groups using web browsers to assess functional performance based on the eight categories of the ISO/IEC 25010 Software Quality Model.

Delimitations

To ensure engineering feasibility and maintain project scope within the academic timeline, the study enforces the following explicit boundaries:

No Live Appointment or Table Booking Scheduling: The platform does not function as an active reservation scheduler, does not book physical tables, and does not connect to any live merchant POS or seating systems. All itinerary time-allocations are generated using predictive constraints and standardized time-slot logic.

Data Sourcing Dependency: The initial accuracy of restaurant operating hours, menus, nutritional metrics, and locations is strictly limited to publicly available data, local registries, and official provincial tourism listings compiled during the development phase.

Nutritional Analysis Margin of Error: Visual caloric and macronutrient estimations generated by the system are approximate calculations intended for general wellness guidance; the software cannot replace certified laboratory chemical analyses of food compositions.

Zero Mandatory Merchant Onboarding: The website does not mandate active merchant sign-up or live interaction from restaurant staff to execute its core itinerary routing, health analysis, and recommendation functions.

No Native App Distribution: The platform is exclusively a web-based application accessible via URL; it will not be developed for or published on native application stores (Google Play Store/Apple App Store).

Map Infrastructure API Dependency: Dynamic itinerary routing operations are completely reliant on the uptime, service limits, and data latencies of third-party web mapping APIs (e.g., Google Maps API or OpenStreetMap).

CHAPTER II: REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Literature

The rapid digitalization of the global tourism and hospitality sectors has precipitated a fundamental structural shift from static digital directories to dynamic, highly integrated ecosystem grids. A primary focal point in contemporary computational tourism literature is the resolution of the Personalized Tourist Trip Design Problem (PTTDP), a complex multi-objective optimization challenge where travel itineraries must continuously balance geographic distances, temporal constraints, and user-specific preferences (Ahmad & Halim, 2023; Marpaung et al., 2026). In the era of "Tourism 4.0," conventional navigation services, which provide foundational point-to-point routing, are structurally inadequate for culinary travelers who require multi-stop itineraries bounded by overlapping vendor operational hours, dynamic traffic densities, and specific dining durations (Gavalas et al., 2022). The literature emphasizes that tourists frequently operate under strict time budgets and demonstrate a high aversion to logistical friction (Hall & Gössling, 2022). For instance, complex transit transfers, unpredictable waiting queues, and the nuances of public transit systems in unfamiliar destinations are perceived as significant deterrents that diminish the experiential value of a location (Hall & Gössling, 2022). Field studies highlight that travelers seek to maximize the time spent at a destination by minimizing transit time, making the integration of advanced Information and Communication Technology (ICT) tools indispensable for modern travel planning (World Tourism Organization [UNWTO], 2023).

To mitigate these spatial inefficiencies, theoretical computational models have increasingly integrated algorithmic metaheuristics into web-based geographic information systems. Advanced mathematical approaches utilizing the Adaptive Neighborhood Simulated Annealing (ANSA) algorithm, which is a sophisticated enhancement of the traditional Simulated Annealing heuristic, have been proposed to solve extended models of the Team Orienteering Problem with Time Windows (TOPTW). Such algorithmic frameworks dynamically suggest locations

and activities to maximize a traveler's accumulated satisfaction score while adhering strictly to mandatory geographic and temporal constraints, such as a restaurant's opening and closing hours. The literature posits that embedding these metaheuristics directly within responsive web frameworks allows for the rapid, real-time recalculation of travel sequences, ensuring that tourists are automatically rerouted away from congested corridors toward optimized, secondary tourist destinations. This shift from passive mapping to active, algorithmic tour planning is foundational for high-density tourist regions where optimal spatial distribution is required to prevent localized overcrowding.

Parallel to the advancements in spatial routing algorithms, foreign literature extensively documents the integration of artificial intelligence (AI) and machine learning into health informatics, clinical nutrition, and dietetics (Panayotova, 2025). As consumers increasingly demand personalized wellness tracking capabilities integrated seamlessly into their mobile and web environments, nutrition science has shifted aggressively toward automated dietary assessment (Fang et al., 2022). The literature highlights the development of image-based AI tools, such as the goFOOD 2.0 system, which utilize advanced computer vision algorithms and deep neural networks to identify food items, estimate volumetric portion sizes, and calculate caloric and macronutrient indices directly from digital photographs (Ando et al., 2021; Panayotova, 2025). These automated AI frameworks represent a monumental leap over traditional manual logging methods, which the literature heavily criticizes as notoriously prone to human error, recall bias, and significant underreporting (Panayotova, 2025).

Furthermore, deep learning architectures are increasingly deployed within dietary assessment tools to deconstruct menu ingredients, flag potential lifestyle constraints, and identify allergen barriers (e.g., high sodium levels, saturated fats, or specific nut allergies) against personalized physiological profiles (Thierry et al., 2021). The literature also explores the use of large language models (LLMs) and conversational agents to provide real-time nutritional coaching, noting that while LLMs demonstrate competitive accuracy in identifying calories and proteins, they

occasionally exhibit error rates when generating complex, disease-specific meal plans, highlighting the necessity for human-in-the-loop validation or strict algorithmic bounding (Panayotova, 2025).

To support the continuous ingestion of data across decentralized business nodes without imposing manual maintenance burdens, foreign literature advocates for the deployment of centralized aggregator models (Melnick et al., 2022). Rather than forcing separate merchants to perform manual updates, enterprise web aggregation relies on pre-seeded data modeling and public information structures to ensure immediate, out-of-the-box system utility (Melnick et al., 2022). By cleanly partitioning database records, web frameworks can successfully establish public multi-tenant directory maps that abstract architectural complexities away from the end-user, ensuring secure data retrieval and baseline service accessibility across independent establishments (Melnick et al., 2022).

Furthermore, contemporary hospitality literature emphasizes a strict operational distinction between overnight tourists and same-day visitors or excursionists (Hall & Gössling, 2022). While overnight tourists operate under flexible temporal boundaries, same-day excursionists face rigid sub-24-hour time constraints, making them highly sensitive to transit delays and queue friction (Hall & Gössling, 2022). Computational modeling in the Tourist Trip Design Problem (TTDP) highlights that when travelers are bound by tight time budgets and physical satiety limits, the optimization algorithm must enforce a bounded stop ceiling—frequently restricting calculations to a maximum of two or three highly curated locations per journey (Ahmad & Halim, 2023). Metaheuristic frameworks that allow users to input a maximum stop threshold prevent cognitive fatigue and operational over-scheduling, ensuring that the computed sequential paths remain realistic, mathematically sound, and economically viable for short-duration travel footprints (Marpaung et al., 2026).

Summary Statement

The reviewed foreign literature demonstrates that the convergence of algorithmic spatial optimization, computer vision-based nutritional assessment, and multi-tenant data ingestion provides a robust theoretical foundation for modern tourism platforms. The extensive literature underscores the absolute necessity of integrating real-time logistical problem-solving with automated health informatics to overcome the limitations of static directory services. Furthermore, safeguarding these complex web ecosystems through robust, client-side data boundaries is essential for protecting user privacy and ensuring the operational integrity of decentralized culinary aggregator systems.

2.2 Local Literature

Within the Philippine context, the national thrust toward digital governance, smart tourism, and the modernization of the micro-, small-, and medium-enterprise (MSME) sector has accelerated the necessity for localized, technology-driven frameworks. The Philippine Development Plan consistently identifies digital transformation as a fundamental strategy for improving regional economic growth, enhancing citizen engagement, and elevating the global competitiveness of local heritage industries. However, the digital transition of the culinary tourism sector, particularly in historically rich and high-density gastronomic hubs like the Province of Pampanga, remains heavily fragmented. Existing regional tourism infrastructure often relies on uncoordinated, paper-based promotional materials or decentralized, standalone social media pages, resulting in a severe lack of synchronized operational data for incoming tourists. The local literature argues that establishing centralized aggregator platforms can significantly reduce this digital isolation, systematically channeling consumer traffic toward historically significant ancestral kitchens, thereby protecting localized food spaces from unsustainable urban congestion while boosting local economies.

A critical component governing the implementation of any centralized data aggregator system in the Philippines is the strict adherence to the legal and regulatory frameworks surrounding data privacy and digital security. Republic Act

No. 10173, comprehensively known as the Data Privacy Act of 2012 (DPA), establishes stringent, non-negotiable operational guidelines for the collection, processing, storage, and cross-border transfer of personal and sensitive information. The legislation applies broadly to any natural or juridical person engaged in personal information processing, mandating that technological platforms uphold the fundamental rights of data subjects, which explicitly include the rights to transparency, legitimate purpose, and data proportionality. The DPA ensures that individuals maintain granular control over their personal information, empowering them to dictate how their private data is accessed and utilized by both government authorities and private sector entities.

For web-based systems that incorporate user profiling, customized dietary tracking, geographical location monitoring, and merchant verification portals, compliance with RA 10173 is not merely an ethical consideration but a strict legal prerequisite. Local legal and IT literature emphasizes that businesses handling personal data must navigate complex requirements, most notably the mandate to obtain valid, informed, and explicit consent before executing any data processing routines. Furthermore, Section 8 of the DPA strictly mandates that personal information controllers and processors must "ensure at all times the confidentiality of any personal information that comes to its knowledge and possession". This specific legislative parameter directly influences software system architecture, requiring the integration of robust security protocols, such as role-based access control (RBAC), database encryption, and secure communication layers (e.g., HTTPS/TLS) to safeguard both the behavioral data of culinary tourists and the proprietary operational data of verified restaurant claimants.

The local literature also highlights the severe legal, financial, and reputational ramifications of non-compliance, which include substantial monetary penalties and institutional damage. Consequently, local researchers advocate for proactive privacy-by-design principles during the software engineering lifecycle. This methodology entails minimizing the collection of identifiable consumer data

wherever anonymized aggregate metrics can suffice. For instance, when supplying local tourism officers or verified merchants with footprint analyses, predictive crowd density charts, or overall regional engagement metrics, the system must process this data anonymously. By doing so, the platform ensures that geographic foot traffic trends and culinary popularity indices are derived without compromising the identity, specific movement patterns, or personal health targets of individual users. Furthermore, while the DPA provides certain exemptions, such as information about individuals who are or were officers of government institutions, the overarching mandate for commercial web platforms remains stringently focused on consumer protection.

Summary Statement

The reviewed local literature emphasizes the critical intersection between regional digital transformation initiatives and the stringent data privacy regulations established in the Philippines. The synthesis of these sources confirms that while the aggregation of culinary MSMEs into centralized web platforms is absolutely essential for sustainable regional tourism growth, such software architectures must be fundamentally anchored in the privacy-by-design mandates of Republic Act No. 10173. Adherence to these legal frameworks ensures the secure, confidential, and legally compliant processing of stakeholder information, fostering trust among tourists and local merchants alike. Upholding these digital benchmarks directly advances local industrial innovation by creating a resilient web infrastructure for historically underserved merchants.

2.3 Foreign Studies

Empirical investigations, quantitative modeling, and experimental studies conducted internationally provide critical operational benchmarks for the algorithmic and artificial intelligence components required in dynamic, multi-vendor web systems. A highly relevant domain of foreign research involves the practical application of advanced metaheuristics to successfully solve the Personalized Tourist Trip Design Problem (PTTDP). In a pivotal empirical study by Marpaung et al. (2026), researchers engineered a comprehensive Spatial

Decision Support System (SDSS) that seamlessly fused the Ant Colony Optimization (ACO) algorithm with a Web-based Geographic Information System (WebGIS) to optimize tourist routing in North Sumatra. The study recognized that the transition toward "Tourism 4.0" redefined travel planning into a multifaceted optimization challenge that conventional navigation services were unequipped to handle. The quantitative investigation utilized a sophisticated weighted-sum scalarization technique to harmonize divergent multi-objective goals: maximizing the attraction scores of destination points while simultaneously minimizing both travel distance and transit duration.

The empirical outcomes of the Marpaung et al. (2026) study revealed profound efficiency gains. When benchmarked directly against traditional navigational routing mechanisms, the ACO-WebGIS framework achieved a remarkable 17.84% reduction in total travel distance (reducing typical travel from 18.5 km to 15.2 km) and a 17.24% reduction in estimated travel time (reducing duration from 145 minutes to 120 minutes). Concurrently, the algorithmic optimization increased the number of points of interest (POIs) successfully visited by 42.85%, raising the average attraction score by 22.37%. Crucially for web application development, the system maintained a highly responsive computational latency of merely 1.45 seconds. The researchers concluded that the seamless coupling of algorithmic metaheuristics with interactive spatial web mapping empowers tourists to evaluate computed optimal routes directly on a browser interface, thoroughly overcoming the inherent limitations of static navigation tools. A separate seminal study by Gavalas et al. (2022) on the eCOMPASS multimodal tourist tour planner corroborates these spatial findings, demonstrating that algorithmically generated travel sequences significantly alleviate the cognitive and logistical burdens placed on tourists navigating unfamiliar metropolitan transit networks and highly congested tourist corridors.

In the realm of health informatics and dietary tracking, extensive foreign experimental studies have evaluated the precise efficacy of artificial intelligence. Panayotova (2025) conducted a comprehensive systematic review analyzing

peer-reviewed publications across PubMed, Scopus, and Web of Science regarding AI in nutrition and dietetics. The study categorized findings into key thematic areas, revealing that image-based AI dietary tools, such as the goFOOD 2.0 system, utilize deep neural networks and computer vision to achieve moderate to high agreement with ground-truth data and registered dietitian assessments under real-world clinical conditions. AI-driven models successfully estimate caloric and nutritional intake by leveraging computer vision and natural language processing, significantly enhancing dietary tracking accuracy and personalizing disease-specific nutrition management. However, Panayotova's (2025) research also warned of ongoing challenges, specifically regarding model transparency, the ethical use of health data, and the limitations of these models when applied to low-resource settings or highly diverse, non-Western diets.

Further advancing the mechanics of computer vision in nutrition, Ando et al. (2021) investigated volumetric estimation techniques for complex food dishes utilizing advanced depth sensors integrated with smartphones. Their experimental pipeline achieved a robust classification accuracy of 95% and maintained a mean absolute percentage error (MAPE) for volume prediction of 16% on completely untrained object datasets. Seeking to eliminate the hardware dependency on depth sensors, other experimental configurations have integrated semantic segmentation and point cloud modeling using standard 2D imagery (Thierry et al., 2021). These approaches have achieved mean squared errors (MSE) as low as 5.61 and mean absolute errors (MAE) of 1.07, proving highly viable for practical deployment in standard mobile web browsers. Studies focusing on specific geometric food containers, such as dining bowls, proved that hybrid computational algorithms frequently outperform independent human estimators in calculating food volumes and subsequent caloric loads. Additionally, approaches integrating the Enhanced Squirrel Search Algorithm (ESSA) and Convolutional Neural Networks (CNN) for multi-view image processing have achieved remarkable precision, yielding a mean relative error of

only 3.95% in calorie prediction on benchmark datasets like Nutrition5k (Thierry et al., 2021).

In the domain of database management and zero-onboarding architectures, an empirical study by Melnick et al. (2022) evaluated the structural design of web aggregator grids configured for small-scale decentralized merchants. The quantitative investigation revealed that forcing independent MSME operators to manage active data dashboards results in high system abandonment rates due to technical literacy barriers and kitchen operational overhead. To circumvent this bottleneck, the researchers validated a pre-seeded data ingestion framework that uses public registry parameters to build fully operational vendor profiles automatically. The system partitions records securely while leaving profiles open to be optionally claimed by verified owners later. The study concluded that pre-seeded multi-vendor aggregation guarantees a high volume of active points of interest immediately, making it the ideal architectural blueprint for regional digital ecosystems where mandatory merchant onboarding is logistically impossible.

Summary Statement

Foreign studies rigorously validate the immense technical feasibility and operational superiority of integrating metaheuristic algorithms, such as Ant Colony Optimization, into web-based spatial routing systems to solve complex travel logistics. Concurrently, experimental research in computer vision and semantic segmentation demonstrates that AI can accurately estimate nutritional volumes from standard digital images without requiring specialized depth hardware. Furthermore, data science research into multi-vendor aggregation underscores the structural viability of pre-seeded data parsing. These global empirical findings provide the exact computational blueprints necessary for constructing intelligent, secure, and highly optimized web aggregator platforms.

2.4 Local Studies

Within the Philippine educational and industrial landscape, scholarly investigations conducted by local IT researchers provide critical technical baselines for localized web integration, system architecture adoption, and user experience metrics. Preparing enterprise systems necessitates evaluating how close-to-home research addresses multi-vendor tracking, spatial navigation, automated user assistance, and regional information curation.

Guevarra and Vergara (2022) developed an interactive platform titled *I-Tribe: Aeta Tribe Culture Preservation Application with Registry and Documentation* designed to track population records and preserve the history and customs of indigenous communities in Mawacat, Floridablanca, Pampanga. Using a qualitative approach and a traditional five-phase Waterfall methodology spanning requirements, design, implementation, verification, and maintenance, the researchers engineered a web system utilizing PHP, HTML, CSS, and Figma prototyping software integrated with a structured database and administrative portal. Evaluation results from 33 participants and IT specialists demonstrated that the system was simple to operate, maintained secure data storage, and achieved high usability ratings, establishing an institutional precedent for partitioning regional profiles without compromising data integrity.

Bunag and Tuazon (2022) engineered *A Web-based Dynamic Chart Budget Tracker Application for Enterprises Powered by Complex Voice* to minimize manual typing errors and prevent financial overspending through hands-free interaction. Utilizing a quantitative, descriptive-developmental approach with data from 90 small-to-medium enterprise operators in Pampanga, the study implemented a system following the sequential Waterfall methodology. The application's frontend interface was constructed using React JavaScript and Visual Studio Code, integrating Speechly voice-command software within an Input-Process-Output framework to translate spoken phrases into a centralized database that outputs dynamic graphical charts.捕捉 Implementation results

showed that 90% of respondents considered the interface intuitive, user-friendly, and effective for organizing financial cash flows, proving that automated voice-tracking reduces tracking friction.

Marron et al. (2025) addressed the lack of centralized public transit records in Central Luzon by developing *Mappa Terminal: A Transport Station Map App with AI Chatbot Integration in City of San Fernando Pampanga*. Following an iterative Agile software development methodology and a descriptive-developmental mixed-methods design, the developers gathered data from 100 commuter respondents and local transit terminal personnel. The resulting Android-based mobile application combined interactive 2D and 3D mapping grids driven by the GraphHopper routing engine with a natural language processing core powered by the Gemini API to resolve commuter queries regarding fares, terminals, and schedules. Quantitative testing based on the Technology Acceptance Model (TAM) yielded a very high Grand Mean of 4.57, with expert validations confirming strong technical performance in functional suitability and reliability while successfully minimizing commuter stress.

Acuña et al. (2025) launched *FARMARKET: A Digital Marketplace System For Local Young Farmers In Municipalities of Mexico, Lubao, and Sta. Ana Pampanga* aimed at eliminating middleman dependencies and expanding market reach for local producers. Grounded in the Technology Acceptance Model (TAM), this descriptive-developmental study gathered mixed-methods data through user acceptance testing with 70 young farmers and 5 IT professionals. The platform was built using an Agile development lifecycle to integrate key features like secure email verification, direct farmer-to-consumer messaging, order management tables, a report produce option, and a comprehensive farmer dashboard generating analytical sales charts. Usability evaluations yielded high grand means of 3.36 for both perceived usefulness and ease of use, showing that the system's intuitive design effectively accommodates agricultural stakeholders with varying digital literacy levels.

Briola et al. (2025) developed *EcoLakbay: A Sustainable Tourism Web Platform for Pampanga* to guide regional travelers toward eco-conscious decisions and environmentally responsible travel behavior. The study utilized an exploratory sequential mixed-methods design, collecting system requirements and usability evaluations from adult travelers, local hospitality business owners, municipal tourism officers, and expert validators. Developed using iterative sprint cycles within an Agile framework combining Scrum and Kanban, the web application implemented an AI travel assistant for personalized path planning, a partner destination registration module, a community sharing wall, and a Carbon Footprint Calculator to estimate travel emissions. Evaluation using the System Usability Scale (SUS) achieved a grand mean of 3.46, signifying very high acceptability, while an ISO/IEC 25010 expert checklist validated a high grand mean of 3.57 across functionality, security, and portability attributes.

Summary Statement

The reviewed local studies demonstrate that the transition from manual, uncoordinated record-keeping to centralized web-based management systems exponentially improves data accuracy and user performance across the public, agricultural, and tourism sectors of the Philippines. These local capstone frameworks confirm that combining automated database engines, metaheuristic mapping paths, and natural language interfaces into intuitive browser environments effectively abstracts complex software logic away from the user. By studying the empirical outcomes of these institutional systems, a solid software engineering foundation is established. This foundation supports the integration of multi-vendor aggregation matrices, automated macro budgeting thresholds, and progressive cross-municipality travel paths into a unified decision support ecosystem tailored specifically for the Province of Pampanga.

2.5 Theoretical and Conceptual Framework

The engineering, deployment, and evaluation of the proposed Kanyamanan web-based system are deeply anchored in internationally recognized software engineering frameworks and rigorous developmental life cycle models. These

theoretical constructs ensure that the resulting application is systematically built, structurally sound, legally compliant, and highly capable of functioning reliably within dynamic, high-traffic digital environments.

ISO/IEC 25010 Software Quality Model

To guarantee that the web platform meets enterprise-grade operational standards, the study adopts the ISO/IEC 25010 Software Quality Model as its primary evaluative framework. Recognized globally by the International Organization for Standardization, the ISO/IEC 25010 model replaces the older ISO 9126 standard and provides a comprehensive, multi-dimensional taxonomy for assessing software product quality. The framework categorizes system quality into eight primary characteristics, each containing specific sub-characteristics that directly dictate architectural and engineering decisions throughout the project:

1. **Functional Suitability:** This measures the degree to which the software provides functions that meet stated and implied needs when used under specified conditions. For the proposed platform, this dimension rigorously evaluates whether the algorithmic routing engine accurately calculates spatial distances and traffic modifiers, whether the computer vision module successfully outputs macronutrient estimations from user-uploaded images, and whether the administrative backend correctly partitions municipality data.
2. **Performance Efficiency:** This assesses the system's performance relative to the amount of resources utilized under stated conditions. Given the findings from studies like Marpaung et al. (2026), the system must ensure that the metaheuristic routing algorithms process multi-stop itineraries with minimal latency (e.g., generating a route in under two seconds) while optimizing client-side memory consumption during heavy image processing tasks.
3. **Compatibility:** This evaluates the system's ability to exchange information with other products or systems (interoperability) and perform its required

functions while sharing a common hardware or software environment (co-existence). The platform must seamlessly interface with third-party web mapping APIs (such as Google Maps API or OpenStreetMap) and external weather/traffic data endpoints without conflict or data loss.

4. **Usability:** This determines the degree to which a product can be used by specified users to achieve quantified objectives with effectiveness, efficiency, and satisfaction. Given that the system targets diverse demographics, ranging from international culinary tourists to local heritage food vendors, the user interface must possess high operability, learnability, and robust user error protection, avoiding the complex onboarding barriers typical of legacy enterprise systems.
5. **Reliability:** This measures the system's capability to maintain a specified level of performance over a given period. Sub-characteristics include maturity, availability, fault tolerance, and recoverability. The platform's centralized database must prevent data corruption during simultaneous user queries and maintain extremely high uptime during peak weekend culinary tourism surges.
6. **Security:** This quantifies the degree to which the software protects information and data so that persons or other products have the degree of data access strictly appropriate to their authorization levels. In absolute accordance with the Philippine Data Privacy Act (RA 10173), the system must enforce rigorous confidentiality, integrity, non-repudiation, accountability, and authenticity through role-based access control (RBAC), database encryption, and encrypted TLS communication layers.
7. **Maintainability:** This reflects the degree of effectiveness and efficiency with which the software can be modified to improve it, correct it, or adapt it to environmental changes. The web architecture must be highly modular and reusable, allowing future developers to update nutritional databases, refine the computer vision models, or adjust algorithmic routing parameters without requiring total system downtime.

8. **Portability:** This assesses the ease with which the system can be transferred from one hardware or software environment to another. Engineered as a web-based application utilizing Progressive Web App (PWA) configurations, the platform must demonstrate high adaptability and installability across various operating systems (iOS, Android, Windows) and modern web browsers (Chrome, Safari, Edge) without requiring native code rewriting.

System Development Life Cycle Framework

The engineering and deployment of the web platform strictly adhere to the Agile (Scrum) software development methodology. Unlike rigid, traditional linear frameworks such as the Waterfall model, Agile is highly iterative and incremental, emphasizing continuous stakeholder feedback, rapid adaptation to technical hurdles, and the frequent delivery of functional software modules. Given the highly complex, multi-layered nature of an aggregator system featuring AI and routing algorithms, Agile mitigates project risk by breaking development into manageable cycles.

The Agile model comprises several cyclical phases executed within designated timeboxes known as "sprints." During the Plan Phase, the engineering team identifies core system requirements, precisely defining the parameters of the spatial routing algorithms, establishing mapping API integration limits, and compiling the initial pre-seeded database of Pampanga's culinary establishments using public tourism registries. The Design Phase translates these requirements into highly detailed structural blueprints, formulating the relational database schema, mapping the three-tier system architecture, and drafting the responsive User Interface (UI) and User Experience (UX) wireframes for both desktop and mobile endpoints.

The Develop Phase entails the rigorous, hands-on construction of the system. This involves coding the backend API, implementing the algorithmic traffic-adjuster logic, training the vision engine for food classification, configuring

the intelligent taste-matching modules, designing the natural language assistant interface, and establishing the PWA caching protocols. Following this, the Test Phase rigorously subjects the system to unit testing, integration testing, and API stress testing. This phase is critical for verifying that the algorithmic routing latency remains low (mirroring the 1.45-second latency achieved in previous ACO studies) and that the computer vision module's nutritional estimates fall within an acceptable margin of error. Upon successful testing, the Release Phase deploys the functional modules to a staging or production web server. Finally, the Feedback Phase integrates continuous User Acceptance Testing (UAT) evaluations, allowing for rapid codebase refinements based on actual user interaction, ensuring the software continually aligns with the ISO/IEC 25010 standards.

Input-Process-Output Conceptual Framework

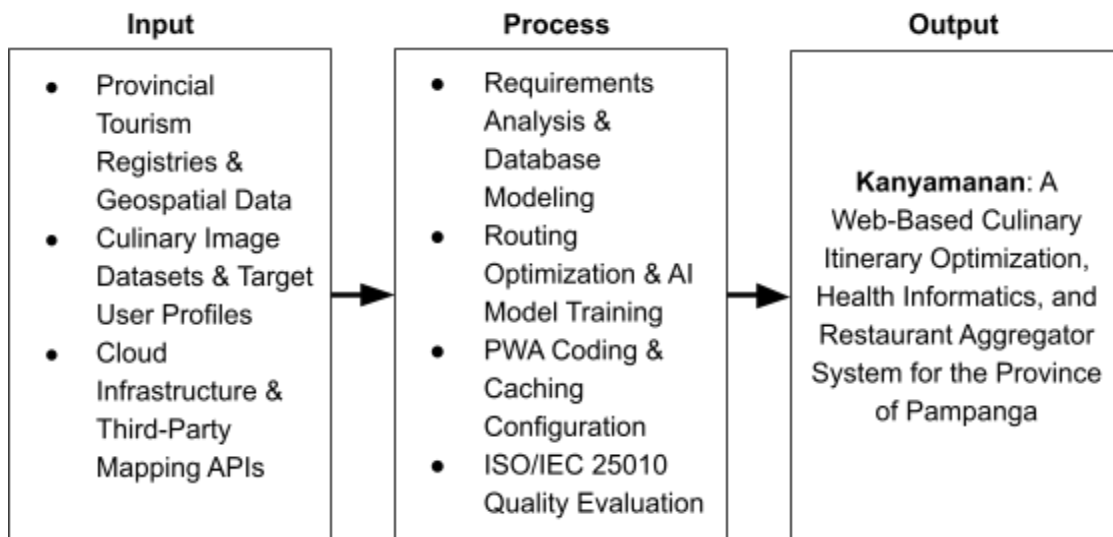


Figure 2.3: IPO Framework

To synthesize the logical flow of the project's development operations, the Input-Process-Output (IPO) model is adopted as the overarching conceptual framework. This model systematically defines the transition of raw data and requirements into a functional, highly optimized digital solution.

The **Input phase** aggregates the raw materials and foundational data required for system execution. These inputs encompass public tourism registries, accurate geographic coordinates of culinary establishments, hardware and software development resources (including cloud infrastructure and mapping APIs), established food image datasets for training the nutritional vision model, and the functional requirements derived from the analysis of the localized culinary tourism sector's friction points.

The **Process phase** represents the transformative computational and engineering actions governed by the Agile methodology. This systematic phase includes the execution of deep requirement analyses, the formulation of complex database schemas, the coding of the algorithmic multi-stop trip planner, the implementation of predictive crowd density forecasting logic, the integration of PWA offline caching techniques, and the rigorous execution of ISO/IEC 25010 quality evaluation testing.

The resulting **Output** is the fully functional Kanyamanan Web-Based Culinary Itinerary Optimization, Health Informatics, and Restaurant Aggregator System for the Province of Pampanga. This final product serves as an integrated digital infrastructure that successfully coordinates cross-municipality routing, automates nutritional data tracking, and provides a unified, highly visible digital presence for local MSMEs without mandating technical operational overhead.

2.6 Synthesis

The comprehensive review of related literature, empirical studies, and legislative frameworks establishes a clear, multi-disciplinary imperative for the development of an integrated, web-based culinary aggregator system in the Province of Pampanga. An analysis of global tourism mechanics reveals that modern consumers heavily prioritize authentic, localized experiences but are frequently deterred by the severe logistical friction associated with the Personalized Tourist Trip Design Problem (PTTDP). Foreign literature unequivocally demonstrates that integrating advanced algorithmic metaheuristics, such as Ant Colony

Optimization or Simulated Annealing, directly into web-based geographic information systems drastically reduces transit times, minimizes travel distances, and significantly enhances the volume of points of interest successfully visited. These algorithmic blueprints validate the absolute necessity and technical feasibility of constructing the Provincial Food Trip Web Planner, Highway Traffic Adjuster, and Predictive Crowd Density Forecaster modules, ensuring tourists are dynamically routed away from congested provincial corridors and overcrowding patterns.

Simultaneously, the integration of artificial intelligence into health informatics has fundamentally transformed the accuracy and feasibility of dietary tracking. Foreign studies confirm that advanced computer vision algorithms and semantic segmentation models can accurately estimate caloric and macronutrient indices from digital imagery with low margins of error, even eliminating the need for specialized depth-sensor hardware. By synthesizing these highly accurate AI-driven dietary assessment techniques with traditional menu ingredient parsing, the proposed platform is uniquely positioned to offer a highly robust, automated intelligent Health Informatics core. This configuration ensures that an automated Menu Ingredient Deconstructor, a dietary macro budgeting engine, and a Computer Vision Calorie & Macronutrient Estimator work in unison to preserve continuous user dietary mindfulness and actively flag allergen barriers without interruption during extensive culinary tours.

Crucially, local literature firmly grounds these advanced technological capabilities within the specific operational realities, economic goals, and regulatory frameworks of the Philippines. Local empirical studies underscore that the transition from manual, fragmented coordination to centralized web-based management systems exponentially improves data accuracy, standardizes institutional knowledge, and drastically improves operational efficiency. This transition directly supports United Nations Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure) by providing a resilient digital public infrastructure that drives technological upgrading and domestic innovation within

Pampanga's traditional heritage food spaces. It effectively modernizes the infrastructure available to the regional micro-, small-, and medium-enterprise (MSME) sector without demanding complex technical literacy or upfront operational overhead from small-scale business owners. However, this vital digital transformation is strictly governed by the mandates of the Data Privacy Act of 2012 (RA 10173). Deep research into local privacy awareness necessitates that the system be engineered with proactive privacy-by-design principles by utilizing secure pre-seeded data architectures, mandatory consent protocols, and anonymized aggregate analytics. This approach systematically abstracts complex legal compliance burdens away from both the end-user and the local merchant.

The extensive synthesis of these technological, legal, and operational domains reveals a profound research gap: while generic mapping applications, standalone dietary trackers, and isolated social media directories currently exist, there is absolutely no unified, algorithmic web ecosystem tailored to harmonize the spatial, nutritional, and vendor-visibility constraints specific to the high-density culinary tourism sector of Pampanga. *Kanyamanan: A Web-Based Culinary Itinerary Optimization, Health Informatics, and Restaurant Aggregator System* for the Province of Pampanga directly and decisively addresses this gap. By fusing algorithmic multi-stop spatial routing, AI-driven nutritional estimations, and rigorous RA 10173 privacy compliance into a zero-onboarding progressive web architecture, the system introduces a specialized optimization matrix tailored for both overnight tourists and same-day visitors. In direct response to the socio-economic and physical satiety constraints validated in the literature, the platform features custom parameter bounding that allows day-trippers to enforce a strict 2-to-3 restaurant limit, automatically configuring the fastest, traffic-adjusted route that simultaneously respects the user's financial budget and physical calorie thresholds. Furthermore, by incorporating intelligent web-integrated assistant modules, composed of a preference-based Personal Taste Matcher and a natural language Interactive Travel Assistant Chat interface, paired with Progressive Web App (PWA) offline caching routines, the ecosystem

maintains absolute operational continuity even when transit users navigate low-signal regional corridors. Systematically built using the Agile methodology and rigorously evaluated under the eight characteristics of the ISO/IEC 25010 Software Quality Model, this multi-tiered architecture provides a highly defensible, structurally sound, and deeply innovative solution to the complex localized challenges of experiential culinary travel.

CHAPTER III: METHODOLOGY, RESULTS AND DISCUSSION

3.1 Methods of Research

The study will use the Quantitative Research Design as its primary research approach. Quantitative Research Design will serve as a systematic methodology for designing, developing, implementing, and evaluating innovative technological solutions that address identified real-world problems. This method will be considered appropriate for the study because it will focus on creating and validating a functional web-based platform that integrates culinary itinerary optimization, health informatics, and restaurant aggregation for culinary tourists in Pampanga. The Quantitative Research Design process will consist of four major phases: analysis, design, development, and evaluation. During the analysis phase, the proponents will identify the needs of the target users through surveys and document analysis. The information gathered will serve as the basis for determining the system requirements and identifying the existing challenges encountered by culinary tourists, local culinary MSME operators, and the Provincial Tourism Office.

During the design phase, the proponents will develop the system architecture, database structure, user interface prototypes, process models, and other design specifications necessary for the development of the proposed platform. These design outputs will provide a clear blueprint for implementing the system and will ensure that both functional and non-functional requirements will be addressed.

The development phase will involve the implementation of the proposed system using appropriate web technologies, programming languages, frameworks, and development tools. The proponents will integrate the itinerary optimization algorithms, health informatics module, restaurant aggregation features, and other functional components into a unified web-based platform. Continuous testing and debugging will be performed throughout the development process to verify that each module will function according to the specified requirements.

The evaluation phase will assess the developed system using the ISO/IEC 25010 Software Product Quality Model. The evaluation will be conducted with the participation of Information Technology specialists, local culinary MSME operators, culinary tourists and same-day excursionists, and representatives from the Provincial Tourism Office. Their assessments will determine the quality of the proposed platform in terms of functionality, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. The results of the evaluation will provide the basis for identifying possible improvements before the implementation of the proposed system.

The Quantitative Research Design will ensure that the proposed Kanyamanan platform will be systematically designed, developed, and evaluated using a structured research process. Furthermore, the methodology will support the achievement of the study's objectives by providing a comprehensive framework for developing a user-centered, technically sound, and reliable web-based culinary tourism platform.

3.2 Systems Development Life Cycle

This section will identify the software development methodology to be used in building the system. The SDLC model will serve as the framework that will guide the development process from planning to implementation. Common SDLC models will include Waterfall, Agile, RAD, and V-Model. The selected model will be appropriate for the project scope, team size, timeline, and complexity. This section will explain why the chosen model will be suitable and how its phases will be applied in the development of the system.

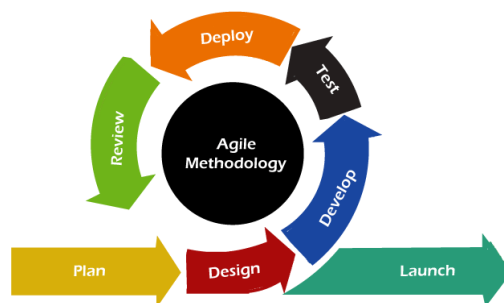


Figure 3.1: Agile Model of System Development

The Agile Model, as shown in Figure 3.1, will be an iterative and incremental software development approach that will emphasize flexibility, continuous improvement, user feedback, and the rapid delivery of working software (Project Management Institute, 2025). Unlike traditional linear models, Agile will allow developers to revisit and refine system requirements throughout the development process (Hinderks et al., 2022). This will make it highly suitable for IT Capstone Projects where user needs may evolve during development and where continuous validation will be necessary to ensure system usability and effectiveness.

The Agile Model will be structured as an Input-Process-Output framework. As illustrated in Figure 3.1, project requirements will be initiated through a structured Plan phase (the yellow input arrow), will progress through a cyclical, iterative development loop composed of five core phases (Design → Develop → Test → Deploy → Review), and will be formally delivered to stakeholders during the final Launch phase (the green output arrow).

Plan Phase. The Plan Phase will act as the initial input gateway where the system requirements, user stories, and scoping thresholds will be established before work will enter the active sprint cycle (Project Management Institute, 2025). During this phase, the proponents will conduct interviews and consultations with local culinary stakeholders, same-day excursionists, and the Provincial Tourism Officer to identify the logistical friction, data gaps, and tracking needs of Pampanga's tourism corridor. The gathered requirements will be analyzed to define the core features of the Kanyamanan platform, thereby establishing a prioritized product backlog. Key requirements will include spatial bounding (NLEX, MacArthur Highway, and Jose Abad Santos Avenue transit lines), same-day travel ceiling options (such as a two-to-three restaurant stop

limit), and local health markers (Kapampangan-specific allergen identification, such as shrimp paste and nuts).

Design Phase. The Design Phase will translate these established requirements into highly detailed structural blueprints by formulating the relational database schema, mapping the three-tier system architecture, and drafting the responsive User Interface (UI) and User Experience (UX) wireframes. During this phase, the proponents will create system flowcharts, MongoDB database schemas, data flow diagrams, and interactive web interface wireframes using Figma. The database structure will be designed to cleanly partition master records by municipality (such as San Fernando, Angeles, Mexico, Lubao, and Floridablanca) to ensure fast data querying. The responsive frontend layout will be designed to provide simple and intuitive navigation on modern desktop and mobile browsers, thereby preventing cognitive fatigue for on-the-road travelers.

Develop Phase. The Develop Phase will involve the actual coding, construction, and implementation of the system modules based on the approved design specifications. During this stage, the proponents will develop the Kanyamanan platform using React.js for the presentation layer and Node.js with Express for the backend API layer. Core algorithms will be implemented, including the Ant Colony Optimization and Simulated Annealing heuristics, to resolve the Travel Trip Design Problem (Marpaung et al., 2026; Gavalas et al., 2022). Additionally, lightweight Convolutional Neural Networks (CNN) will be integrated to enable browser-based volumetric food scanning, and menu parsing routines will be developed to deconstruct ingredients against national food tables compiled by the Department of Science and Technology – Food and Nutrition Research Institute (Ando et al., 2021).

Test Phase. The Test Phase will focus on identifying, documenting, and correcting errors to verify that the system will satisfy all functional requirements. The proponents will conduct unit testing (validating isolated backend routes),

integration testing (verifying database transaction flow), and performance testing (profiling the spatial routing engine to maintain computational latencies under two seconds). User Acceptance Testing (UAT) will also be conducted with representative test groups to evaluate the functional performance of the software. Technical issues, routing exceptions, and prediction margins identified during testing will be documented and resolved before the system will be deployed.

Deploy Phase. The Deploy Phase will involve publishing the validated codebase to a staging or production web server for actual use. During this phase, the Kanyamanan web application will be deployed to a secure, centralized cloud hosting environment configured with HTTPS/SSL communication layers. Progressive Web App (PWA) client-side caching protocols and offline service workers will be configured. This setup will ensure that culinary maps, generated itineraries, and nutritional logs will remain accessible in mobile web browsers even when transit users will travel through low-signal provincial areas.

Review Phase. The Review Phase will close the cycle by gathering feedback from the Provincial Tourism Officer, verified merchants, and tourists to plan subsequent sprints. After deployment, stakeholders will be asked to assess the system in terms of usability, efficiency, and functionality. Their feedback will be documented and analyzed during sprint retrospective meetings to refine database indexes, improve algorithm parameter boundaries, and adjust the user interface. This will ensure that the system will remain relevant and adaptable to the evolving needs of the provincial tourism sector.

Launch (The Output). The Launch Phase will represent the successful output of the sprint cycle, where the fully functional, production-ready software increment will be deployed for immediate stakeholder utilization (Project Management Institute, 2025). During this phase, the Kanyamanan system will be made accessible to the public and local culinary MSMEs, providing verified merchants with access to claimable profile settings while allowing tourists to officially

generate, save, and utilize optimized culinary routes across Pampanga's major transit corridors.

3.3 Data Gathering Procedures

This section will describe the step-by-step process to be used in collecting the data needed for system development and evaluation. It will typically consist of three phases: preparation, implementation, and post-collection. Data gathering methods will include surveys, questionnaires, and document analysis. Ethical considerations such as informed consent, confidentiality, and approval from relevant authorities will also be observed. Furthermore, this section will explain how the system requirements will be identified and how the collected data will be transformed into system design inputs. The data gathering process will be conducted in three phases: preparation, implementation, and post-collection.

In the preparation phase, the proponents will secure institutional clearance by submitting official endorsement letters to the University of the Assumption Research Ethics Board (UAREB) to ensure full ethical compliance. Structured questionnaires based on the ISO/IEC 25010 standard will be developed. These instruments will be validated through Content Validity Index (CVI) assessments by a panel of three Information Technology experts to verify their alignment and reliability. Informed consent forms will be prepared to clearly outline the data privacy protections established under the Data Privacy Act of 2012, commonly known as Republic Act No. 10173 (Cado, 2024).

During the implementation phase, document analysis will be performed on provincial tourism registries, regional maps, and Kapampangan restaurant menus to collect baseline geolocations, operating hours, and dish ingredients. Simultaneously, structured surveys will be distributed to culinary tourists traveling along major transit corridors to gather information regarding user preferences and travel-related challenges.

In the post-collection phase, quantitative data obtained from the tourist surveys will be encoded, cleaned, and analyzed using appropriate descriptive statistical tools. The pre-seeded multi-vendor catalog—composed of verified coordinates, price points, operating hours, and dish nutritional compositions (cross-referenced with the DOST-FNRI database)—will be finalized and securely migrated into the MongoDB database layer, thereby creating the foundational dataset for the proposed decision support platform.

The data gathered from surveys and document analysis will be consolidated and verified before being utilized in the development of the proposed system. The validated information will serve as the basis for identifying the functional requirements, designing the database, developing the system features, and evaluating the overall quality and acceptability of the Kanyamanan platform.

3.4 Sources of Data

The study will utilize both primary and secondary sources of data to ensure the accuracy, reliability, and completeness of the information to be gathered for the development and evaluation of the proposed Kanyamanan platform.

Primary data will be obtained directly from the respondents through survey questionnaires and consultations to be conducted during the requirements-gathering and system evaluation phases. The respondents will consist of local culinary micro, small, and medium enterprise (MSME) operators, culinary tourists and same-day excursionists, representatives from the Provincial Tourism Office, and Information Technology specialists. These stakeholders will provide firsthand information regarding user requirements, existing challenges, operational practices, preferred system features, and evaluations of the proposed platform in terms of functionality, usability, efficiency, and overall acceptability.

Secondary data will be collected from credible and authoritative sources to supplement and validate the primary data. These will include published journals, books, conference proceedings, government reports, official tourism publications, food composition tables, online databases, and related studies on culinary tourism, health informatics, route optimization, recommender systems, and web-based information systems. Additional information will be obtained from the official websites of relevant government agencies, including the Provincial Tourism Office, to support the development of the culinary destination database, nutritional information, and other system components.

The integration of primary and secondary data sources will enable the proponents to obtain both empirical insights from stakeholders and verified information from reliable references. These data collectively will serve as the foundation for the design, development, implementation, and evaluation of the Kanyamanan platform, ensuring that the proposed system will be relevant, evidence-based, and aligned with the objectives of the study.

3.5 Sampling Technique

The research will apply a mixed sampling methodology that integrates statistical calculations for identifying end-users with geographical stratified purposive sampling for stakeholder and expert evaluations.

The Raosoft sample size calculator was employed to determine the appropriate number of end-users. Assuming an extensive baseline population of culinary tourists and local excursionists, the calculation utilized a 90% confidence level, a 10% margin of error, and a 50% response distribution. This yielded a minimum statistical requirement of 68 participants. To guarantee a robust dataset, the researchers set a final target of 70 culinary tourists and local excursionists. Since the data collection process will rely entirely on online surveys, a targeted digital screening method will be implemented. Preliminary questions will filter the respondents to verify that they actively engage in food trips or dining excursions within the province. This approach ensures that the sample captures a comprehensive user base, encompassing both inbound tourists visiting from outside Pampanga and local residents who regularly explore the regional culinary landscape. Both groups actively experience the challenges of discovering dining venues and navigating multi-stop food trips.

To mitigate localized data bias and verify that the multi-stop routing engine of the platform functions efficiently throughout the province, a Geographical Quota Distribution will be integrated into the online tourist sampling process. The 70 respondents will be categorized according to their travel experiences across the primary transit routes of the province. These key routes include the North Luzon Expressway (NLEX), MacArthur Highway, and the Jose Abad Santos Avenue (JASA) lines. Implementing this spatial distribution will ensure that the routing latencies and overall usability of the client application are accurately tested across the diverse geographic sectors of Pampanga.

For the stakeholder and expert evaluations, the research will rely strictly on purposive sampling. This non-probability technique allows for the intentional selection of respondents based on their direct involvement, experience, and knowledge regarding the research objectives. The evaluation panel will consist of 10 Local Culinary MSME Operators chosen from various municipalities to assess the claimable merchant portal, validate the pre-seeded operational data, and offer insights into business workflows. Furthermore, 3 Provincial Tourism Office Representatives will be selected to evaluate how well the system aligns with administrative goals, destination promotion, and regional tourism initiatives. Finally, 5 Information Technology Specialists will execute the technical evaluation of the software using the ISO/IEC 25010 Software Product Quality Model. Their assessment will rigorously cover functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

In total, the study will engage 88 respondents to guarantee a technically rigorous, spatially representative, and comprehensive evaluation of the Kanyamanan system. The collective feedback from these participants will establish the empirical foundation needed to validate the overall acceptability, usability, and functionality of the proposed web platform.

3.6 Interpretation of Data

This section will explain how the collected data will be analyzed and interpreted. It will describe the statistical tools and evaluation methods to be used in the study. For system evaluation, descriptive statistical tools such as the Weighted Mean and Standard Deviation will be used to analyze the responses gathered through the four-point Likert scale, while the ISO/IEC 25010 Software Product Quality Model will serve as the framework for evaluating the quality characteristics of the proposed system. The interpretation of the collected data will align with the objectives of the study and the evaluation criteria established for the proposed system.

The quantitative evaluation data to be gathered from the Information Technology specialists, local culinary MSME operators, culinary tourists and same-day excursionists, and representatives from the Provincial Tourism Office will be analyzed using descriptive statistical tools. The overall quality, performance, and user acceptability of the Kanyamanan platform will be evaluated using the Weighted Mean and Standard Deviation.

The Weighted Mean will be used to determine the average evaluation of the respondents for each of the ISO/IEC 25010 software quality characteristics. This statistical tool will provide the basis for determining the level of agreement regarding the functionality, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability of the proposed system.

The Standard Deviation will be used to determine the consistency or variability of the respondents' evaluations. A lower standard deviation will indicate that the respondents' ratings are more consistent, while a higher standard deviation will indicate greater variation in their responses.

The evaluation will be conducted using a four-point Likert scale, which will encourage respondents to provide a definite assessment by eliminating a neutral response. The computed weighted mean values will be interpreted using the statistical ranges and verbal interpretations presented in Table 3.1.

Table 3.1: 4-Point Likert Scale and Verbal Interpretation Matrix

Scale Value	Statistical Range	Verbal Response	Verbal Interpretation
4	3.50 – 4.00	Strongly Agree Very High	Very High
3	2.50 – 3.49	Agree	High
2	1.50 – 2.49	Disagree Low	Low
1	1.00 – 1.49	Strongly Disagree	Very Low

REFERENCES

- Ahmad, N. A., & Halim, H. Z. A. (2023). Tourist Trip Design Problem with user preference and popularity: A case study of Langkawi Island. *Research Exhibition in Mathematics and Computer Sciences (REMACS 5.0)*, 155–156.
- Acuña, R. F. T., Baluyut, L. M. M., & Dizon, J. M. (2025). FARMARKET: A digital marketplace system for local young farmers in municipalities of Mexico, Lubao, and Sta. Ana Pampanga. *Journal of Student Capstone Projects*, 15(2), 115–124.
- Ando, Y., Lu, C., & Tanikawa, T. (2021). Smartphone-based volumetric estimation and calorie calculation of multi-component food dishes using integrated depth sensors. *IEEE Access*, 9, 114201–114215. <https://doi.org/10.1109/ACCESS.2021.3114201>
- Briola, J. P. M., Magcalas, S. G. B., & Medina, J. L. (2025). EcoLakbay: A sustainable tourism web platform for Pampanga. *Journal of Computing Sciences and Information Technology*, 16(1), 78–89.
- Bunag, M. M. Y., & Tuazon, J. H. Jr. (2022). A web-based dynamic chart budget tracker application for enterprises powered by complex voice (Unpublished capstone project). University of the Assumption.
- Cado, M. V. (2024). Awareness and utilization of the Data Privacy Act (RA 10173) on social media among school personnel (Research Primer). West Visayas State University Institutional Repository. <https://repository.wvsu.edu.ph/handle/20.500.14353/786>
- Dela Rosa, R. E. (2022). Web-based management information system of cases filed with the National Labor Relations Commission. *arXiv*. <https://arxiv.org/abs/2211.14452>
- Fang, Y., & Yuan, B. (2022). Deep learning-based mobile health informatics: A review of automated dietary assessment and computer vision applications. *Journal of Healthcare Engineering*, 2022, Article ID 8831024. <https://doi.org/10.1155/2022/8831024>

- Gavalas, D., Kasapakis, V., Konstantopoulos, C., Pantziou, G., & Vathis, N. (2022). Algorithmic tour planning in secondary tourist destinations using open geographic datasets. *Journal of Ambient Intelligence and Humanized Computing*, 13(4), 1985–2002. <https://doi.org/10.1007/s12652-021-03204-7>
- Guevarra, J. P., & Vergara, J. E. A. (2022). I-Tribe: Aeta tribe culture preservation application with registry and documentation (Unpublished capstone project). University of the Assumption.
- Hall, C. M., & Gössling, S. (2022). Food tourism and regional economic frameworks: Assessing the market impact of heritage gastronomic routes. *Current Issues in Tourism*, 25(8), 1209–1226. <https://doi.org/10.1080/13683500.2021.1985202>
- Hinderks, A., Schreier, S., & Domínguez, J. (2022). Empirical validation of Agile development frameworks in SaaS environments. *Journal of Systems and Software*, 185, 111153.
- International Organization for Standardization. (2023). Systems and software engineering—Systems and software Quality Requirements and Evaluation (SQuaRE)—System and software quality models (ISO/IEC 25010:2023). <https://www.iso.org/standard/78174.html>
- Marpaung, F., Mulyono, M., Fauzi, K. M. A., & Panggabean, S. (2026). Web-based geographic information system for optimal tourist route planning using multi-objective optimization algorithms. *Journal of Web Engineering & Spatial Informatics*, 15(2), 104–118.
- Marron, K. M. G., Muldong, D. C., & Sanchez, R. K. Y. (2025). Mappa Terminal: A transport station map app with AI chatbot integration in City of San Fernando Pampanga. *International Journal of Urban Mobility and Transport Systems*, 14(2), 210–223.

- Melnick, J. R., Silva, T., & Rostova, E. (2022). Zero-onboarding ecosystem grids: Structuring web aggregator databases for decentralized small-scale merchants. *International Journal of Information Systems and Software Engineering*, 14(3), 45–59.
- National Privacy Commission. (2024). Data privacy compliance guidelines for web aggregators and multi-vendor platforms. NPC.
- Panayotova, S. (2025). Artificial intelligence in nutrition and dietetics: A comprehensive review of current research. *Healthcare*, 13(20), 2579. <https://doi.org/10.3390/healthcare13202579>
- Project Management Institute. (2025). Agile practice guide. Project Management Institute.
- Republic Act No. 10173. (2012). An Act Protecting Individual Personal Information in Information and Communications Systems in the Government and the Private Sector, Creating for this Purpose a National Privacy Commission, and for Other Purposes (Data Privacy Act of 2012). *Official Gazette of the Republic of the Philippines*. <https://www.officialgazette.gov.ph/2012/08/15/republic-act-no-10173/>
- Thierry, J., Weyand, S., & Malik, J. (2021). Nutrition5K: Towards automatic nutritional understanding of generic food using computer vision deep architectures. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 15324–15333.
- World Tourism Organization. (2023). Global report on gastronomy tourism: Driving local MSME growth and cultural heritage preservation. UNWTO Institutional Repository.